3.(a) // STACK USING ARRAY PROGRAM

```
#include<stdio.h>
int stack[100],n,top,i;
void push(int x)
{
   if(top>=n-1)
   {
      printf("\n STACK is over flow");
   }
   else
   {
      top++;
      stack[top]=x;
      printf("\n element pushed into stack:%d",x);
   }
}
void pop()
{
   if(top<=-1)
   {
      printf("\n Stack is under flow");
   }
   else
   {
      printf("\n The popped elements is %d",stack[top]);
      stack[top] = 0;
      top--;
   }
}
void display()
{
   if(top>=0)
   {
      printf("\n The elements in STACK: \n");
      for(i=top; i>=0; i--)
         printf("%d \n",stack[i]);
   }
   else
   {
      printf("\n The STACK is empty");
   }

}
```

```
int main()
{
	top = -1;
	n = 5;
	push(10);
	push(20);
	push(30);
	push(40);
	push(50);
	push(60);

	display();
	pop();
	pop();
	pop();
	pop();
	pop();
	pop();
	pop();
}
```